This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

Performance Analysis of Wireless Energy Harvesting Cognitive Radio Networks Under Smart Jamming Attacks

Dinh Thai Hoang[1], Dusit Niyato[1], Ping Wang[1], and Dong In Kim[2]
[1] School of Computer Engineering, Nanyang Technological University (NTU), Singapore
[2] School of Information and Communication Engineering, Sungkyunkwan University (SKKU), Korea

***Abstract*—In cognitive radio networks with wireless energy harvesting, secondary users are able to harvest energy from a wireless power source and then use the harvested energy to transmit data opportunistically on an idle channel allocated to primary users. Such networks have become more common due to pervasiveness of wireless charging, improving the performance of the secondary users. However, in such networks, the secondary users can be vulnerable to jamming attacks by malicious users who can also harvest wireless energy to launch the attacks. In this paper, we first formulate the throughput optimization problem for a secondary user under the attacks by jammers as a Markov decision process (MDP). We then introduce a new solution based on the deception tactic to deal with smart jamming attacks. Furthermore, we propose a learning algorithm for the secondary user to find an optimal transmission policy and extend to the case with multiple secondary users in the same environment. Through the simulations, we demonstrate that the proposed learning algorithms can effectively reduce adverse effects from smart jammers even when they use different attack strategies.**

***Index Terms*—Cognitive radio, wireless energy harvesting, jamming attack, deception strategy.**

I. INTRODUCTION

Cognitive radio networks (CRNs) with wireless energy harvesting have been receiving growing attention recently. By integrating cognitive radio with wireless power transfer technology, unlicensed users, i.e., secondary users, are not only allowed to access licensed channels opportunistically, but also supplied with energy wirelessly for their transmission. In particular, in CRNs, secondary users can use the channels allocated for licensed users, i.e., primary users, when the channels are not occupied by primary users [1]. However, energy supply for the secondary users is still limited. The development of wireless power transfer techniques can be adopted to solve the energy problems caused by the constrained hardware of mobile devices. With wireless power transfer, mobile nodes can work in a fully untethered fashion without supports from a wired power source or battery replacement.

Although wireless energy harvesting CRNs can solve spectrum resource management and power supply problems, such kind of networks can be vulnerable to malicious attacks especially for secondary users. In CRNs, secondary users do not own licensed spectrum and thus they cannot be protected from the attacks by adversaries. Moreover, with wireless power supply, adversaries can also harvest the energy to perform the jamming attacks to the data transmissions of the secondary users with the aim to degrade the network performance. In this paper, we consider that the jammers are capable of distinguishing between the signals from primary users and that from secondary users by using smart sensing processes.

The classical solutions to jamming attacks in wireless systems often assume that there are multiple channels and jammers cannot attack all channels simultaneously. Therefore, secondary users can choose some channels which are unlikely to be attacked or they will switch to another channel if the current channel is jammed by the jammers. However, in many cases, secondary users are designed to transmit data opportunistically on a dedicated channel, for example, when only the licensed user of that channel allows to do so. Although jammers can attack secondary users on a single channel, they also have limited energy harvested from wireless energy sources, and they may not be able to perform attacks continuously. Therefore, to deal with the jamming attacks on the target channel, we propose a novel solution that is developed based on the deception tactic in military. The main idea of the deception mechanism is using fake transmissions to undermine the attack ability of enemies, e.g., by wasting the energy of their adversaries. Thus, jammers may not be able to attack when secondary users transmit actual information.

We first consider the wireless energy harvesting cognitive radio networks with a secondary user and multiple smart jammers. Both the secondary user and jammers are able to harvest energy from a wireless power source, e.g., nearby wireless chargers. For the jammers, they will use the harvested energy to perform attacks by jamming into a target channel. Likewise, the secondary user uses this harvested energy to transmit data or to perform deceptions to undermine jammers. To find an optimal strategy for the secondary user, i.e., transmit data or perform deception, we first formulate the throughput optimization problem for the secondary user under the attacks by the jammers as a Markov decision process (MDP). Then, a learning algorithm based on simulation-based methods is proposed to help the secondary user cope with the jammers when the secondary user has no information about jammers, e.g., the number of jammers, and their strategies. Additionally,

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIP) (2014R1A5A1011478), Singapore MOE Tier 1 (RG18/13 and RG33/12) and MOE Tier 2 (MOE2014-T2-2-015 ARC 4/15).

Dinh Thai Hoang, Dusit Niyato, and Ping Wang are with Nanyang Technological University, Singapore. E-mails: dinh0007@e.ntu.edu.sg, dniyato@ntu.edu.sg, wangping@ntu.edu.sg.

Dong In Kim is with Sungkyunkwan University, Korea. E-mail: dikim@skku.ac.kr. Dong In Kim is the corresponding author.

the secondary user can adapt to the dynamics of cognitive networks, i.e., when the secondary user has no information about primary users, e.g., the idle channel, miss detection, false alarm, and successful transmission probabilities. Through the learning process, the secondary user can adjust its policy to maximize throughput in an online fashion. Then, we extend the optimization to the case with multiple secondary users coexisting in the same environment. We propose learning algorithms to help secondary users achieve their objective of throughput maximization. Finally, we perform simulations to demonstrate the efficiency as well as the convergence of the proposed learning algorithms. Simulation results show that with multiple jammers and with different strategies, the learning algorithms can dynamically adjust data transmission and deception actions to find the optimal solutions for secondary users. We provide convergence proofs and comparisons with other algorithms.

Note that we assume that the smart jammers aim to degrade the performance of the data transmission by secondary users only. The primary users are not their targets. For example, there is a regulator of licensed users to apply severe penalty to the jammers. However, the optimization model can still be extended for primary users which can be studied in a future work.

The rest of the paper is organized as follows. In Section II, a comprehensive review of the related work is given. Section III describes the system model together with assumptions considered in this paper. In Section IV, we consider the case with single secondary user and propose a learning algorithm to find the optimal policy for the secondary user. Section V extends to the case with multiple secondary users. Finally, performance evaluation results are presented in Section VI, and we conclude the paper in Section VII.

II. Related Work

A. Cognitive Radio Networks with Wireless Energy Harvesting

The performance optimization problem for secondary users in cognitive radio networks (CRNs) with wireless energy harvesting was studied in the literature. In [5], Sultan *et al.* studied the throughput optimization problem for a secondary user in a CRN. The secondary user is assumed to be able to harvest energy from an environment and it uses the harvested energy to transmit data on a target channel when the channel is idle. The authors then formulated the optimization problem as an MDP and used the stochastic dynamic programming method to obtain the optimal policy. The throughput optimization problem for secondary users was also considered in [6]. However, instead of finding an optimal policy through the dynamic programming method as presented in [5], the authors in [6] examined a suboptimal solution to reduce the complexity. Different from [5] and [6] which find an optimal energy usage policy, in [7], Park *et al.* aimed to find an optimal detection threshold to maximize the throughput for secondary users under the causality and collision constraints. One of the most important conclusions is that when the energy arrival rate is lower than the expected energy consumption, the number of channel access attempts is limited by the amount of harvested energy. In [8], a framework for cooperative communications with polarization signal processing technique was introduced to avoid interference between primary and secondary users. Additionally, similar to [5], the MDP framework and the dynamic programming solution were used to maximize the throughput for secondary users.

Recently, there are some research works proposed to exploit wireless energy for secondary users in CRNs [9], [10], [11]. In CRNs with radio frequency (RF) harvesting, if a channel is idle, secondary users can opportunistically transmit data. By contrast, if the channel is busy, they can harvest energy through using RF energy harvesting techniques [14], [15] and use the harvested energy for future data transmissions. Therefore, in [9] and [10], Park *et al.* considered the throughput optimization problems for secondary users through the partially observable Markov decision processes (POMDP) framework. There is a slight difference in formulating the optimization problem between [9] and [10]. While in [9] the authors aimed to find an optimal policy for the secondary user to access a channel or to harvest energy, in [10], an optimal policy is to determine whether to sense the channel or remain idle. While in both [9], [10], Park *et al.* considered only one target channel, in [11], [12], Hoang *et al.* proposed a learning solution based on the simulation-based method to deal with multi-channel sensing problem. The simulation results showed that the performance in terms of throughput of the secondary user obtained by the proposed learning algorithm is close to that of the optimization solutions.

Nevertheless, there are some arguments that RF energy harvesting may not be sufficient to support an operation of a secondary user. Therefore, in this paper, we consider a more general case that the secondary user and jammers can receive energy from wireless charging, e.g., inductive coupling or magnetic resonance coupling [13]. Therefore, channel access and wireless energy harvesting can be performed concurrently.

B. Jamming Attacks in Cognitive Radio Networks

Although, in CRNs, secondary users benefit from opportunistic channel access, they are vulnerable to jamming attacks from malicious users. Therefore, a few solutions were proposed to address this problem. The most commonly used strategy is frequency hopping. The main idea of this solution relies on the assumption that there are multiple orthogonal or non-overlapping channels so that secondary users can choose a channel which is unlikely to be attacked or they can switch to the other channel if the current channel is being attacked. For example, in [16], Li *et al.* first modeled the interaction between a secondary user and a jammer using game theory. The action of the jammer is to choose channels which the secondary user may use. The action of the secondary user is to select channels which are not attacked by jammers. For one-shot game, the authors proved that the game has a unique Nash equilibrium. The authors then considered the cases with multiple secondary users and with multistage game and discuss the Nash equilibria for both cases. In addition, in [17] the authors extended [16] by using the adversarial bandit algorithm [18] to address the unknown channel statistics

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

problem which was not considered in [16]. In both papers, numerical results demonstrated the efficiency of the proposed algorithms. A similar game model was also proposed in [19] and [21]. However, in [19], Sodagari *et al.* proposed another method using multi-armed bandit strategies [20], while in [21] Wu *et al.* developed an MDP-based defense strategy which is approximately close to the game equilibrium. To achieve more efficient spectrum usage, the authors in [22] proposed using dedicated control channels for exchanging control information among secondary users. Therefore, the action of a secondary user is now to determine the number of control and data channels with the aim to maximize the throughput of the secondary user. Through using minimax-Q learning to obtain the optimal policy for the secondary user, the authors showed that the achievable performance of the system is much better than that of the myopic learning algorithm.

Besides channel hopping strategies, there are also some other strategies used to deal with jamming attacks in CRNs. In [23], the authors proposed an idea to avoid jamming attacks through *proxy* users. Specifically, it is assumed that there are N available channels at the user side and there are other M channels at the base station side. When the secondary users cannot connect directly with the base station, they can use relay nodes, i.e., proxy users, to transmit data to the base station. In this way, jammers need to jam both proxies and followers, and thus the attack efficiency will be reduced. Consequently, the throughput of secondary users will be improved. In [24], an anti-jamming strategy based on honeypot was proposed. In this strategy, secondary users will be selected to be a honeynode in a round-robin fashion. When the honeynode transmits data to one of channels and the channel is attacked by jammers, instead of hopping to other channels, it will accept the sacrifice by continuously transmitting to attract jammers and thus other users will have more opportunities to transmit data successfully. Another method, called power control strategy, was proposed in [25] to overcome the jamming attack problem. In [25], the authors assumed that a secondary user is equipped with multi-antenna and thus it can transmit data on different channels at the same time. The secondary user aims to maximize its throughput, while jammers want to minimize successful transmissions of the secondary user. Then, a Colonel Blotto game [26] was used to model the interaction between the secondary user and jammers. Through simulation, the authors demonstrated that to minimize the damage caused by jammers, the secondary user should use the minimax strategy [27].

In summary, in all above works and others in the literature, to the best of our knowledge, the performance optimization problem for secondary users in a cognitive radio network with wireless energy harvesting under jamming attacks has still not yet been studied. Furthermore, none of them investigated using deception strategy in dealing with jamming attacks in such environment, e.g., with only one target channel. There is also no work adopting an online learning algorithm with simulation-based methods to find optimal strategies for secondary users. They are the main goals of this paper.

III. System Model and Assumptions

We consider a cognitive radio network with energy harvesting capability as shown in Fig. 1. There are secondary users (SUs) and jammers coexisting in the same environment. While the secondary users use overlay spectrum access technique [1] to avoid interferences with primary users' signals when they want to transmit data to their destinations on a cognitive radio channel, the jammers aim to impede the communication by jamming attacks. The primary user uses the channel to transmit data in a time slot basis and the channel idle probability is denoted by $p_{\mathrm{c}}^{\mathrm{idle}}$. At the beginning of each time slot, the SU senses the channel. If the channel is busy, the SU will do nothing. Nevertheless, if the channel is idle, the SU will transmit data on the channel if it has enough energy. We also consider the spectrum sensing errors of the SU for two cases, namely, miss detection and false alarm. The miss detection happens when the current channel is busy, but the sensing result of the SU is idle. The false alarm occurs when the present channel is idle, but the sensing result is busy. The miss detection and false alarm probabilities of SUs are denoted by $p_{\mathrm{su}}^{\mathrm{miss}}$ and $p_{\mathrm{su}}^{\mathrm{false}}$, respectively.

Each SU has a data buffer to store incoming data and a battery to store energy harvested from an energy source through wireless energy harvesting. The harvested energy then will be used to transmit data in the data queue of SUs. We assume that each SU can harvest energy and transmit data simultaneously. For example, the SU receives energy from a wireless charger through its charging interface. The data queue of the SU has two states, i.e., idle or busy, corresponding to when the SU has no data and when the SU has data to transmit, respectively. The capacity of the battery of the SU is E_{su} units of energy and in each time slot, the user can harvest $e_{\mathrm{su}}^{\mathrm{har}}$ units of energy with probability $\lambda_{\mathrm{su}}^{e}$. When the SU has no energy, it will do nothing. When the SU has enough energy, it can choose to do nothing, to transmit data if the data queue is not empty, or to perform a deception action by transmitting a fake packet on the channel to trap the jammers. When the SU transmits data or a fake packet on the channel, if the jammer detects signals from the SUs and it has enough energy, it will perform an attack by jamming into the channel for the entire time slot. We assume that the jammers are able to distinguish the signals transmitted by the primary users and secondary users.

In practice, there are some techniques, e.g., by cyclostationary feature detection or link signatures, as presented in [1], [2] to detect and recognize the signal from the primary user. In particular, in [1] (Chapter 2), the matched filter detection could be the solution for jammers to detect and attack secondary users' signals. In this method, if the jammers have information about the signal from a primary user, they can use this information as a template. Then, if the signal received on the channel is not matched with the template, it will be treated as secondary users' signals and the jammers will jam to the secondary users' transmissions. This technique is suitable when the transmission of primary users has pilot, preambles, synchronization word or spreading codes, which can be used to construct the template for spectrum sensing. However, with

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

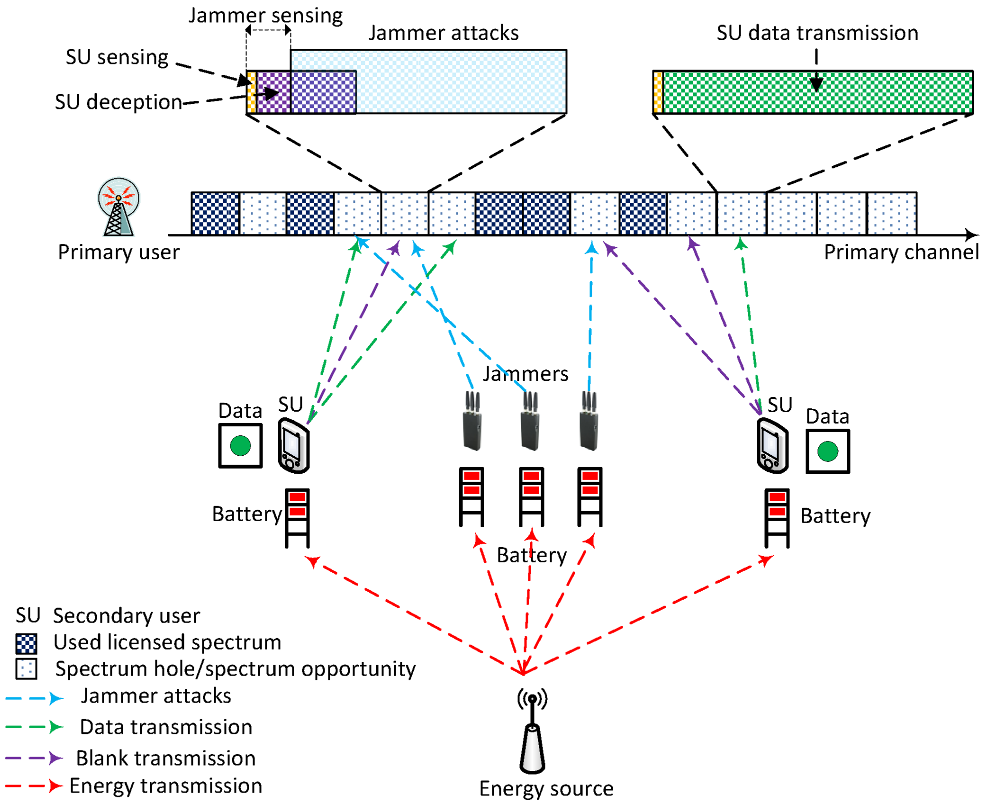


Fig. 1. System model.

the matched filter detection technique, the secondary users can mimic primary users by transmitting imitated signals on the channel in order to cheat the jammers. In this way, the jammers can use many other techniques introduced in the literature to detect the signals of the secondary users. These techniques are known as the primary user emulation detection problem in cognitive radio networks [2], [3], [4].

The main goal of this paper is to find a solution that maximizes the throughput for secondary users given the attacks by the jammers. Here, there is the tradeoff between transmitting data and performing deception. If the SU performs the deception by transmitting a fake packet, it can possibly deceive the jammers to attack and weaken them by wasting their energy. Hence, when the SU transmits actual data, the jammers may not have enough energy to attack. We assume that the SU requires $e_{\mathrm{su}}^{\mathrm{dec}}$ units of energy to perform the deception, and it requires $e_{\mathrm{su}}^{\mathrm{tran}}$ units of energy to transmit actual data. In practice, we have $e_{\mathrm{su}}^{\mathrm{dec}} < e_{\mathrm{su}}^{\mathrm{tran}}$. This can be, for example, that the SU transmits only fraction of a time slot to deceive the jammers and also save its energy. This is illustrated in Fig. 1.

The jammers are equipped with a battery with capacity of E_{ja} units of energy. Similar to the SUs, in each time slot, the jammers can harvest $e_{\mathrm{ja}}^{\mathrm{har}}$ units of energy with probability $\lambda_{\mathrm{ja}}^{e}$. At the beginning of each time slot, the jammers perform channel sensing. If they detect signals from SUs and have enough energy, they will perform attacks by jamming transmission signals. We assume that a jammer has to use $e_{\mathrm{ja}}^{\mathrm{att}}$ to perform the attack. In practice, the jamming signal strength, which is referred to as PHY layer jamming, is measured by the jamming-to-signal ratio [28] defined as follows:

$$\frac{J}{S} = \frac{P_j G_{jr} G_{rj} R_{tr}^2 L_r B_r}{P_t G_{tr} G_{rt} R_{jr}^2 L_j B_j} \tag{1}$$

where j, r, and t represent the jammer, receiver and transmitter, respectively. P_x is the transmission power of a node x ($x \in \{j, r, t\}$), G_{xy} is the antenna gain from node x to node y ($y \in \{j, r, t\}$), R_{xy} is the distance between node x and node y, L_x is the signal loss of the communication link, and B_x is bandwidth. From (1), to perform the jamming attack successfully, the jammer needs to have a high jamming-to-signal ratio and this can be achieved through controlling parameters such as P_j, and R_{jr}. Controlling R_{jr} is difficult since if the jammer comes close to the receiver (i.e., reduce R_{jr} to increase $\frac{J}{S}$), the receiver will detect the jammer easily and punish the jammer. By contrast, controlling the jamming power is an easier way and is commonly used in jamming attacks. There are some existing works that consider the jamming power control for jammers as presented in the survey [29]. However, this is not main focus of this paper. In this paper, we focus on the secondary users' strategies to deal with jamming attacks. Furthermore, we assume that to jam the channel successfully, the jamming power is higher than data transmission power, i.e., $e_{\mathrm{ja}}^{\mathrm{att}} \geq e_{\mathrm{su}}^{\mathrm{tran}}$.

There are J jammers in the network and they have two strategies to attack SUs, called, independent and coordinated attacks. For the independent attack strategy, the jammers perform the attacks separately without knowing about other jammers and thus it is possible that multiple jammers could jam the same channel at the same time. By contrast, for the coordinated attack strategy, the jammers can communicate and one of them is selected to attack the channel at a time. In this paper, we assume that the jammers with enough energy will be randomly chosen to perform the attack to the transmission of the SU. Thus, the other jammers can conserve their energy for the future attacks.

In the following, we will propose the solutions for the SUs to deal with the attacks from jammers with the aim to maximize the throughput of SUs. We will study two cases, i.e., single and multiple SUs. In the first case, there is only one SU, and we propose a learning algorithm for the SU to find an optimal data transmission and deception policy. Then, in the second case, there are multiple secondary users. A similar learning algorithm will be adopted that manages multiple access and deception.

IV. An Optimal Policy for Single Secondary User

In this section, we consider the case of a single secondary user (SU) under the attacks from jammers. We first formulate an optimization problem for the SU as an MDP. Then, we propose a learning algorithm based on the policy gradient method to find an optimal policy for the SU.

A. Problem Formulation

1) State Space: We define the state space of the SU as follows:

$$\mathcal{S} \triangleq \Big\{(d,e); d \in \{0,1\}, e \in \{0,1,\ldots,E_{\mathrm{su}}\}\Big\}, \qquad (2)$$

where d represents the number of packets in the data queue, i.e., data queue state, and e represents the energy level of the battery of the SU. Without loss of generality, for the number of packets, there are two values, i.e., 1 and 0 for having or not having data to transmit, respectively. An extension for arbitrary size of the data queue is straightforward. For the battery, the energy level is divided into discrete levels.

2) Action Space: At each decision epoch, the SU needs to make a decision after it has channel sensing result. There are three actions for the SU to choose from, namely, do nothing, perform deception, or transmit data. Thus we can define the action space as follows:

$$\mathcal{A} \triangleq \Big\{a : a \in \{1,2,3\}\Big\}, \qquad (3)$$

where

$$a = \begin{cases} 1, & \text{when the SU does nothing} \\ 2, & \text{when the SU performs deception,} \\ & \text{i.e., transmitting a fake packet} \\ 3, & \text{when the SU transmits actual data} \end{cases}$$

Additionally, the action space given the state of the SU $\mathcal{A}_s$ comprises all possible actions that do not make a transition to the state that is not allowed. We can express the action space $\mathcal{A}_s$ at the current state $s \in \mathcal{S}$ as follows:

$$\mathcal{A}_s = \begin{cases} \{1\}, & \text{if } e_s < e_{\mathrm{su}}^{\mathrm{dec}}, \\ \{1,2\}, & \text{if } e_{\mathrm{su}}^{\mathrm{dec}} \le e_s < e_{\mathrm{su}}^{\mathrm{tran}} \\ & \text{OR if } e_{\mathrm{su}}^{\mathrm{tran}} \le e_s \text{ and } d_s = 0, \\ \{1,2,3\}, & \text{if } e_{\mathrm{su}}^{\mathrm{tran}} \le e_s \text{ and } d_s > 0. \end{cases} \qquad (4)$$

The first condition corresponds to the case when the battery is almost empty. The SU can select only action $a = 1$, i.e., do nothing. The second condition corresponds to the case that the SU has enough energy only for performing the deception. The SU can select to do nothing or transmit a fake signal. The third condition corresponds to the case that the SU has enough energy to transmit actual data.

3) Transition Probability: For a conventional MDP, we need to construct a transition probability matrix. However, in many cases including the system model under our consideration, the construction of the transition probability matrix is intractable and impossible. For example, to derive the transition probability matrix for the SU, we need complete and perfect information about the networks, e.g., the channel idle probability $p_{\mathrm{c}}^{\mathrm{idle}}$, the miss detection and false alarm probabilities, i.e., $p_{\mathrm{su}}^{\mathrm{miss}}$ and $p_{\mathrm{su}}^{\mathrm{false}}$, and the information about jammers, e.g., the number of jammers and their strategies. Since the objective of the jammers is to attack and impair secondary users' transmissions, and thus they will never disclose their information. Consequently, deriving the transition probability matrix for the SU is infeasible in this case. Therefore, in this paper, we propose a learning algorithm that is developed based on a simulation-based method [30], [33]. The main idea of the simulation-based method is using a "simulator" that can simulate the interaction among the SU, cognitive radio channel, and jammers by generating parameters (e.g., idle channel, miss detection, and false alarm probabilities) of a given system. From the simulation process, the SU can update its information, and then make optimal decisions accordingly.

In the simulation-based method, for a given control policy Ψ, the joint transition probability function $\mathbf{p}$ can be expressed as follows:

$$\mathbf{p}\big(s(t+1)|s(t),\Psi\big) = \mathbf{p}\Big[\big(d(t+1),e(t+1)\big)|\big(d(t),e(t)\big),\Psi\Big]$$
$$= \begin{cases} \mathbf{p}_{\mathrm{sim}} p\Big(\big(d(t),e(t)\big)\Big) p(\Psi(a(t))), & \text{if } d(t+1) = \mathcal{D}^* \\ & \text{and } e(t+1) = \mathcal{E}^* \\ 0, & \text{otherwise} \end{cases} \qquad (5)$$

where $\mathbf{p}_{\mathrm{sim}}$ is a probability function of the simulator that generates simulation parameters for the system process, e.g., parameters of the cognitive radio channel and jammers. $p\Big(\big(d(t),e(t)\big)\Big)$ is the probability that the SU is at state s which is represented by the number of packets d and energy level e at time slot t. $p(\Psi(a(t)))$ is the probability that the SU takes action a under the control policy Ψ at time slot t. Moreover, $\mathcal{D}^* = \min\Big(\big([d(t) - \phi(t)]^+ + \lambda(t)\big), 1\Big)$, and $\mathcal{E}^* = \min\Big(\big[e(t) - c(t)]^+ + y(t)\big], E_{\mathrm{su}}\Big)$. Again, E_{su} is the maximum capacity of the battery of the SU. Here, $\phi(t)$ is the number of packets transmitted at time slot t, $\lambda(t)$ is the number of arriving packets, $y(t)$ is the amount of energy harvested, and $c(t)$ is the amount of energy used at time slot t. Furthermore, $[x]^+ = \max(x,0)$.

4) Reward Function: We define the immediate reward function for the SU as the throughput as follows:

$$\mathscr{T}(s,a) = \begin{cases} 1, & \text{if a packet is successfully transmitted} \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

B. Parameterization for the MDP

We denote $\Theta = \{\theta_{s,a} \in \mathbb{R}\}$ as a parameter vector of the SU at state s given the current action a. We consider a randomized parameterized policy [30], [31], [32] to find the decision for the SU. Under the randomized parameterized policy, when the SU is at state s, it will choose action a with the probability $\mu_\Theta(s,a)$ which is obtained as follows:

$$\mu_\Theta(s,a) = \frac{\exp\left(\theta_{s,a}\right)}{\sum_{a_i \in \mathcal{A}_s} \exp\left(\theta_{s,a_i}\right)}. \tag{7}$$

Note that every $\mu_\Theta(s,a)$ must not be negative and

$$\sum_{a \in \mathcal{A}_s} \mu_\Theta(s,a) = 1. \tag{8}$$

Under the randomized parameterized policy $\mu_\Theta(s,a)$, the transition probability function and the immediate reward function can be parameterized as follows:

$$\mathbf{p}(s'|s,\Psi(\Theta)) = \sum_{a \in \mathcal{A}_s} \mu_\Theta(s,a)\mathbf{p}(s'|s,a), \tag{9}$$

$$\mathscr{T}(s,\Theta) = \sum_{a \in \mathcal{A}_s} \mu_\Theta(s,a)\mathscr{T}(s,a), \tag{10}$$

for all $s, s' \in \mathcal{S}$.

In this paper, we aim to maximize the average throughput of the SU under the randomized parameterized policy $\mu_\Theta(s,a)$ which is expressed as follows:

$$\psi(\Theta) = \lim_{t\to\infty} \frac{1}{t}\mathbb{E}_\Theta\Big[\sum_{k=0}^{t} \mathscr{T}(s_k,\Theta_k)\Big], \tag{11}$$

where s_k is the state of the SU at time slot k. $\mathbb{E}_\Theta[\cdot]$ is the expectation under parameter vector Θ.

We make the following assumption:

Assumption 1. *The Markov chain is aperiodic and there exists a state s^* which is recurrent for each of such a Markov chain.*

Assumption 1 states that the system possesses a Markov property, and we always can find a state s^* such that it will be revisited with a positive probability. Under Assumption 1, the average throughput $\psi(\Theta)$ does not depend on the initial parameter vector Θ_0, and it is well defined on every Θ. Moreover, with Assumption 1, we have following balance equations:

$$\begin{aligned} &\sum_{s \in \mathcal{S}} \pi_\Theta(s)\mathbf{p}(s'|s,\Psi(\Theta)) = \pi_\Theta(s'),\ \text{for } s' \in \mathcal{S}, \\ &\sum_{s \in \mathcal{S}} \pi_\Theta(s) = 1, \end{aligned} \tag{12}$$

where $\pi_\Theta(s)$ is the steady-state probability at state s under the parameter vector Θ. These balance equations have a unique solution defined as a vector $\Pi_\Theta = \begin{bmatrix} \cdots & \pi_\Theta(s) & \cdots \end{bmatrix}^\top$. With steady-state probability, the average throughput can be defined by an alternative way as follows:

$$\psi(\Theta) = \sum_{s \in \mathcal{S}} \pi_\Theta(s)\mathscr{T}(s,\Theta). \tag{13}$$

C. An Idealized Gradient Method

As presented in [34], to update the parameter vector Θ, we can use a gradient method with a standard form as follows:

$$\Theta_{k+1} = \Theta_k + \rho_k \nabla\psi(\Theta_k), \tag{14}$$

where ρ_k is a step size and $\nabla\psi(\Theta_k)$ is the gradient of average throughput.

We then make the following assumptions for the policy gradient method.

Assumption 2. *For every state $s, s' \in \mathcal{S}$, the transition probability $\mathbf{p}(s'|s,\Psi(\Theta))$ and the immediate throughput function $\mathscr{T}(s,\Theta)$ are bounded, twice differentiable, and have bounded first and second derivatives.*

Assumption 3. *The step size ρ_k is deterministic, nonnegative and satisfies the following conditions:*

$$\sum_{k=1}^{\infty} \rho_k = \infty,\ \textit{and}\ \sum_{k=1}^{\infty} (\rho_k)^2 < \infty. \tag{15}$$

Assumption 2 indicates that the transition probability function and the immediate reward function depend smoothly on Θ. This assumption is important when we apply gradient methods for adjusting Θ. Assumption 3 ensures to meet the necessary conditions for convergence of the policy gradient method. Under Assumption 1 and Assumption 3, it is proved in [34] that $\lim_{k\to\infty} \nabla\psi(\Theta_k) = 0$ and thus $\psi(\Theta_k)$ converges.

To determine the gradient of average throughput $\nabla\psi(\Theta_k)$ in Equation (14), we introduce Proposition 1 as follows:

Proposition 1. *Let Assumption 1 and Assumption 2 hold, then*

$$\begin{aligned} &\nabla\psi(\Theta) = \\ &\sum_{s \in \mathcal{S}} \pi_\Theta(s)\Big(\nabla\mathscr{T}(s,\Theta) + \sum_{s' \in \mathcal{S}} \nabla\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta)\Big), \end{aligned} \tag{16}$$

where $d(s',\Theta)$ is the differential throughput at state s'. In general, we can define the differential throughput at state s as follows:

$$d(s,\Theta) = \mathbb{E}_\Theta\left[\sum_{k=0}^{T-1} \left(\mathscr{T}(s_k,\Theta) - \psi(\Theta)\right)|s_0 = s\right], \tag{17}$$

where $T = \min\{k > 0 | s_k = s^*\}$ is the first future time that state s^* is visited. Here, we note that the main aim of defining the differential throughput $d(s,\Theta)$ is to represent a relation between the average throughput and the immediate throughput at state s, instead of the recurrent state s^*. Additionally, under Assumption 1, the differential throughput $d(s,\Theta)$ is a unique solution of the Bellman equation defined as follows:

$$d(s,\Theta) = \mathscr{T}(s,\Theta) - \psi(\Theta) + \sum_{s' \in \mathcal{S}} \mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta), \tag{18}$$

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

for all $s \in \mathcal{S}$.

Proposition 1 establishes the gradient of the average throughput $\psi(\Theta)$. The proof of Proposition 1 is provided in Appendix A.

D. Learning Algorithm

With the idealized gradient method, we can maximize the average throughput of the SU if we can determine the gradient of the average throughput function $\psi(\Theta)$ with respect to Θ. Then, we update the value of the parameter vector Θ in each time slot as presented in Equation (14). However, if the system has a large state space, it is difficult to calculate the value of $\nabla\psi(\Theta)$. Thus, we consider an alternative method which can estimate the value of $\nabla\psi(\Theta)$ and update the value of Θ in an online fashion.

From (8), we have $\sum_{a\in\mathcal{A}_s}\mu_\Theta(s,a) = 1$. Thus, we can derive that $\sum_{a\in\mathcal{A}_s}\nabla\mu_\Theta(s,a) = 0$ for every Θ. From (10), we have

$$\begin{aligned}\nabla\mathscr{T}(s,\Theta) &= \sum_{a\in\mathcal{A}_s}\nabla\mu_\Theta(s,a)\mathscr{T}(s,a), \quad (19)\\ &= \sum_{a\in\mathcal{A}_s}\nabla\mu_\Theta(s,a)(\mathscr{T}(s,a)-\psi(\Theta)), \quad (20)\end{aligned}$$

Moreover, from (9) we have

$$\begin{aligned}&\sum_{s'\in\mathcal{S}}\nabla\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta)\\ &=\sum_{s'\in\mathcal{S}}\sum_{a\in\mathcal{A}_s}\nabla\mu_\Theta(s,a)\mathbf{p}(s'|s,a)d(s',\Theta)\end{aligned} \quad (21)$$

for all $s \in \mathcal{S}$.

Therefore, along with Proposition 1, we can derive the gradient of $\psi(\Theta)$ as follows:

$$\begin{aligned}\nabla\psi(\Theta) &= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\Big(\nabla\mathscr{T}(s,\Theta)+\\ &\sum_{s'\in\mathcal{S}}\nabla\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta)\Big),\\ &=\sum_{s\in\mathcal{S}}\pi_\Theta(s)\Big(\sum_{a\in\mathcal{A}_s}\nabla\mu_\Theta(s,a)(\mathscr{T}(s,a)-\psi(\Theta)))\\ &+\sum_{s'\in\mathcal{S}}\sum_{a\in\mathcal{A}_s}\nabla\mu_\Theta(s,a)\mathbf{p}(s'|s,a)d(s',\Theta)\Big),\\ &=\sum_{s\in\mathcal{S}}\pi_\Theta(s)\sum_{a\in\mathcal{A}_s}\nabla\mu_\Theta(s,a)\Big((\mathscr{T}(s,a)-\psi(\Theta))\\ &+\sum_{s'\in\mathcal{S}}\mathbf{p}(s'|s,a)d(s',\Theta)\Big),\\ &=\sum_{s\in\mathcal{S}}\sum_{a\in\mathcal{A}_s}\pi_\Theta(s)\nabla\mu_\Theta(s,a)q_\Theta(s,a),\end{aligned}$$

where

$$\begin{aligned}q_\Theta(s,a) &= \Big(\mathscr{T}(s,a)-\psi(\Theta)\Big)+\sum_{s'\in\mathcal{S}}\mathbf{p}(s'|s,a)d(s',\Theta),\\ &=\mathbb{E}_\Theta\left[\sum_{k=0}^{T-1}\left(\mathscr{T}(s_k,a_k)-\psi(\Theta)\right)|s_0=s,a_0=a\right].\end{aligned} \quad (22)$$

Here s_k and a_k are the state and action at time slot k, respectively, and $T = \min\{k > 0|s_k = s^*\}$ is the first future time that the current state s^* is visited. $q_\Theta(s,a)$ can be interpreted as the differential throughput if action a is taken based on policy μ_Θ at state s. Then, we present Algorithm 1 that updates the parameter vector Θ at the visit to the recurrent state s^*.

Algorithm 1 Algorithm to update parameter vector Θ at the visit to the recurrent state s^*

At the time slot k_{m+1} of the $(m+1)$th visit to state s^*, we update the parameter vector Θ and the estimated average throughput $\widetilde{\psi}$ as follows:

$$\Theta_{m+1} = \Theta_m + \rho_m F_m(\Theta_m, \widetilde{\psi}_m), \quad (23)$$

$$\widetilde{\psi}_{m+1} = \widetilde{\psi}_m + \kappa\rho_m \sum_{k'=k_m}^{k_{m+1}-1}\Big(\mathscr{T}(s_{k'},a_{k'})-\widetilde{\psi}_m\Big), \quad (24)$$

where

$$F_m(\Theta_m,\widetilde{\psi}_m) = \sum_{k'=k_m}^{k_{m+1}-1}\widetilde{q}_{\Theta_m}(s_{k'},a_{k'})\frac{\nabla\mu_{\Theta_m}(s_{k'},a_{k'})}{\mu_{\Theta_m}(s_{k'},a_{k'})}, \quad (25)$$

$$\widetilde{q}_{\Phi_m}(s_{k'},a_{k'}) = \sum_{k=k'}^{k_{m+1}-1}\Big(\mathscr{T}(s_k,a_k)-\widetilde{\psi}_m\Big). \quad (26)$$

In Algorithm 1, κ is a positive constant and ρ_m is a step size that satisfies Assumption 3. The term $F_m(\Theta_m,\widetilde{\psi}_m)$ represents the estimated gradient of the average throughput. It is calculated by the cumulative sum of the total estimated gradient of the average throughput between two successive visits (i.e., mth and $(m+1)$th visits) to the recurrent state s^*. Furthermore, $\nabla\mu_{\Theta_m}(s_{k'},a_{k'})$ expresses the gradient of the randomized parameterized policy function that is provided in (7). Algorithm 1 enables us to update the parameter vector Θ and the estimated average throughput $\widetilde{\psi}$ iteratively. Accordingly, we derive the following convergence result for Algorithm 1.

Proposition 2. *Let Assumption 1 and Assumption 2 hold, and let $(\Theta_0,\Theta_1,\ldots,\Theta_\infty)$ be the sequence of the parameter vectors generated by Algorithm 1 with a suitable step size ρ satisfying Assumption 3, then $\psi(\Theta_m)$ converges and*

$$\lim_{m\to\infty}\nabla\psi(\Theta_m) = 0, \quad (27)$$

with probability one.

The proof of Proposition 2 is given in Appendix B.

In Algorithm 1, to update the value of the parameter vector Θ at the next visit to the state s^*, we need to store all values of $\widetilde{q}_{\Theta_m}(s_k,a_k)$ and $\frac{\nabla\mu_{\Theta_m}(s_k,a_k)}{\mu_{\Theta_m}(s_k,a_k)}$ between two successive visits. However, this method could result in slow processing. Therefore, we modify Algorithm 1 to improve the efficiency.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

First, we rewrite $F_m(\Theta_m, \widetilde{\psi}_m)$ as follows:

$$\begin{aligned} F_m(\Theta_m, \widetilde{\psi}_m) &= \sum_{k'=k_m}^{k_{m+1}-1} \widetilde{q}_{\Theta_m}(s_{k'}, a_{k'}) \frac{\nabla \mu_{\Theta_m}(s_{k'}, a_{k'})}{\mu_{\Theta_m}(s_{k'}, a_{k'})} \\ &= \sum_{k'=k_m}^{k_{m+1}-1} \frac{\nabla \mu_{\Theta_m}(s_{k'}, a_{k'})}{\mu_{\Theta_m}(s_{k'}, a_{k'})} \sum_{k=k'}^{k_{m+1}-1} \Big(\mathscr{T}(s_k, a_k) - \widetilde{\psi}_m \Big) \\ &= \sum_{k'=k_m}^{k_{m+1}-1} \big(\mathscr{T}(s_k, a_k) - \widetilde{\psi}_m \big) z_{k+1}, \end{aligned} \quad (28)$$

where

$$z_{k+1} = \begin{cases} \frac{\nabla \mu_{\Theta_m}(s_k, a_k)}{\mu_{\Theta_m}(s_k, a_k)}, & \text{if } k = k_m, \\ z_k + \frac{\nabla \mu_{\Theta_m}(s_k, a_k)}{\mu_{\Theta_m}(s_k, a_k)}, & k = k_m + 1, \dots, k_{m+1} - 1. \end{cases} \quad (29)$$

The detail of the algorithm is given in Algorithm 2, where κ is a positive constant and ρ_k is the step size of the algorithm.

Algorithm 2 Algorithm to update Θ at every time slot

At time slot k, the state is s_k, and the values of Θ_k, z_k, and $\widetilde{\psi}(\Theta_k)$ are available from the preceded iteration. We update z_k, Θ_k, and $\widetilde{\psi}_k$ according to:

$$z_{k+1} = \begin{cases} \frac{\nabla \mu_{\Theta_k}(s_k, a_k)}{\mu_{\Theta_k}(s_k, a_k)}, & \text{if } s_k = s^*, \\ z_k + \frac{\nabla \mu_{\Theta_k}(s_k, a_k)}{\mu_{\Theta_k}(s_k, a_k)}, & \text{otherwise}, \end{cases} \quad (30)$$

$$\Theta_{k+1} = \Theta_k + \rho_k (\mathscr{T}(s_k, a_k) - \widetilde{\psi}_k) z_{k+1}, \quad (31)$$

$$\widetilde{\psi}_{k+1} = \widetilde{\psi}_k + \kappa \rho_k (\mathscr{T}(s_k, a_k) - \widetilde{\psi}_k). \quad (32)$$

With Algorithm 2, in each time slot, each SU just needs to update two parameter vectors, i.e., Θ, and $\widetilde{\psi}$, efficiently with simple calculations.

V. Optimal Policy for Multiple Secondary Users

In this section, we consider the case with multiple secondary users coexisting in the same environment. They are coordinated to achieve the highest cooperative throughput from channel access. We first propose a centralized learning solution and then introduce a learning algorithm based on time division multiple access (TDMA) for the cooperation among SUs.

A. Centralized Learning Solution

In this case, we assume that there exists a centralized node, e.g., an access point, that collects information from SUs, i.e., data queue state and energy level, and then makes decisions for SUs.

1) State Space: There are N SUs in the network and at the beginning of each time slot, the SUs send their data queue state and energy level information to the access point. Thus, the global state space of the system now can be defined from local state spaces of SUs as follows:

$$\mathcal{S}^{\text{cen}} \triangleq (\mathcal{S}^1 \times \cdots \times \mathcal{S}^n \times \cdots \times \mathcal{S}^N), \quad (33)$$

where $\mathcal{S}^n$ is the local state space of SU-n and it is defined as in (2), i.e.,

$$\mathcal{S}^n \triangleq \Big\{ (d^n, e^n); d^n \in \{0, 1\}, e^n \in \{0, 1, \dots, E^n_{\text{su}}\} \Big\}. \quad (34)$$

2) Action Space: Similarly, the joint action space $\mathcal{A}^{\text{cen}}$ is a composition of local action spaces of all SUs. The joint action space can be defined as follows:

$$\mathcal{A}^{\text{cen}} \triangleq (\mathcal{A}^1 \times \cdots \times \mathcal{A}^n \times \cdots \times \mathcal{A}^N), \quad (35)$$

where $\mathcal{A}^n$ is defined as in (3), i.e.,

$$\mathcal{A}^n \triangleq \Big\{ a^n : a^n \in \{1, 2, 3\} \Big\}. \quad (36)$$

Since there is only one channel, we need to impose a constraint for the joint action space such that there is at most one SU using the channel at a time. The SU may transmit actual data or a fake packet. Consequently, the joint action $a^{\text{cen}} = (a^1, \dots, a^n) \in \mathcal{A}^{\text{cen}}$ must ensure that not only local action spaces given the states of SUs, i.e., $\mathcal{A}^n_{s^n}\ \forall n \in \{1, \dots, N\}$, comprise all possible actions which do not result in transition into the state that is not allowed as shown in (4), but also there is no collision by SUs on the channel. With the constraint, the joint action space can be determined by Proposition 3 as follows:

Proposition 3. *At time slot t, the system is at the state $s^{\text{cen}} = (s^1, \dots, s^N)$ and the action space corresponding to the current state is $\mathcal{A}^{\text{cen}}_{s^{\text{cen}}} = (\mathcal{A}^1_{s^1}, \dots, \mathcal{A}^N_{s^N})$. Then, we have*

$$|\mathcal{A}^{\text{cen}}_{s^{\text{cen}}}| = \sum_{n=1}^{N} |\mathcal{A}^n_{s^n}| - N + 1, \quad (37)$$

where $|\mathcal{A}^n_{s^n}|$ is the cardinality of the action space $\mathcal{A}^n$ of SU-n given the current state s_n.

The proof of Proposition 3 is given in Appendix C. From Proposition 3, when all SUs have the action space, i.e., $|\mathcal{A}^n_{s^n}| = 3\ \forall n \in \{1, \dots, N\}$, the bound of action space of the system is $2N + 1$ which significantly reduces the size of the action space.

Then, the transition probability function and the immediate reward function can be defined in a similar way as in Section IV as follows: and

$$\mathscr{T}^{\text{cen}}(s^{\text{cen}}, a^{\text{cen}}) = \begin{cases} 1, & \text{if a packet is successfully transmitted}, \\ 0 & \text{otherwise}. \end{cases} \quad (39)$$

After formulating the optimization problem, we can use the learning algorithm based on a simulation method as presented in Section IV to find optimal policies for SUs. The details of the learning algorithm is presented in Algorithm 3.

B. TDMA-based Learning Solution

Learning algorithms with the support from an access point can obtain optimal policies for SUs. However, the communication between SUs and the access point can be expensive because of information exchange in every time slot. Furthermore, even with the proposed constraints, the state space is still large. Therefore, in this section, we adopt the time division multiple access (TDMA) mechanism combining with the learning algorithms introduced in Section IV. Compared with the proposed centralized learning algorithm, by using the TDMA mechanism, we can avoid the communication

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

$$\begin{aligned} &\mathbf{p}\big(s^{\mathrm{cen}}(t+1)|s^{\mathrm{cen}}(t),\Psi^{\mathrm{cen}}\big) = \mathbf{p}\Big[\big(d^{\mathrm{cen}}(t+1),e^{\mathrm{cen}}(t+1)\big)|\big(d^{\mathrm{cen}}(t),e^{\mathrm{cen}}(t)\big),\Psi^{\mathrm{cen}}\Big] \\ &= \begin{cases} \mathbf{p}_{\mathrm{sim}}\Pi_{n=1}^{N} p\Big(\big(d^n(t),e^n(t)\big)\Big)p(\Psi^{\mathrm{cen}}(a^n(t))), & \text{if } d^n(t+1)=\mathcal{D}_n^* \text{ and } e^n(t+1)=\mathcal{E}_n^* \\ 0, & \text{otherwise,} \end{cases} \end{aligned} \tag{38}$$

$$\begin{aligned} &\mathbf{p}\big(s^{n}(t+1)|s^{n}(t),\Psi^{n}\big) = \mathbf{p}\Big[\big(d^{n}(t+1),e^{n}(t+1)\big)|\big(d^{n}(t),e^{n}(t)\big),\Psi^{n}\Big] \\ &= \begin{cases} (\mathbf{p}_{\mathrm{sim}})^N p\Big(\big(d^n(t),e^n(t)\big)\Big)p(\Psi(a^n(t))), & \text{if } d^n(t+1)=\mathcal{D}_n^* \text{ and } e^n(t+1)=\mathcal{E}_n^*, \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \tag{43}$$

Algorithm 3 The centralized learning algorithm

Require: Secondary users establish connections to the access point.

for $t := 1$ to T **do**

SUs send their information, i.e., data queue state and energy level, along with sensing results to the access point.

The access point makes decisions based on information received from secondary users and sends the decisions to secondary users.

The secondary users perform actions received from the access point and they send the results back to the access point.

From the received results, the access point updates the global system values, i.e., z^{sys}, Θ^{sys}, and $\widetilde{\psi}^{sys}$ according to:

$$z_{k+1}^{sys} = \begin{cases} \frac{\nabla \mu^{sys}_{\Theta_k^{sys}}(s_k^{sys},a_k^{sys})}{\mu^{sys}_{\Theta_k^{sys}}(s_k^{sys},a_k^{sys})}, & \text{if } s_k^{sys}=s^*, \\ z_k^{sys} + \frac{\nabla \mu^{sys}_{\Theta_k^{sys}}(s_k^{sys},a_k^{sys})}{\mu^{sys}_{\Theta_k^{sys}}(s_k^{sys},a_k^{sys})}, & \text{otherwise,} \end{cases} \tag{40}$$

$$\Theta_{k+1}^{sys} = \Theta_k^{sys} + \rho_k(\mathscr{T}^{sys}(s_k^{sys},a_k^{sys}) - \widetilde{\psi}_k^{sys})z_{k+1}^{sys}, \tag{41}$$

$$\widetilde{\psi}_{k+1}^{sys} = \widetilde{\psi}_k^{sys} + \kappa\rho_k(\mathscr{T}^{sys}(s_k^{sys},a_k^{sys}) - \widetilde{\psi}_k^{sys}). \tag{42}$$

SUs update information of the data queue and energy level based on environment parameters, e.g., arriving packets and energy harvested.

end for

overhead and reduce the complexity as well as evade the curse of dimensionality in state space and action space. With the TDMA mechanism, the optimization problem for SUs is similar to that presented in Section IV because there is only one SU accessing the channel in a time slot. The difference is at the transition probability function for SU-n as follows: Here, $(\mathbf{p}_{\mathrm{sim}})^N$ must be calculated N times because in TDMA the SUs make decisions for using the channel in a round-robin fashion. In particular, the SUs still need to update the state from the data queue and energy level.

After formulating the throughput optimization problem for SUs, we can apply the learning algorithm in Section IV to find optimal policies for SUs in the network as shown in Algorithm 4. The proof of the convergence of Algorithm 4 is similar to the proof of the convergence of Algorithm 1 which is provided in Appendix B.

Algorithm 4 The TDMA-learning algorithm

Require: Each secondary user determines its time sequence number, i.e., the time slot that it is allowed to access the channel in a round-robin fashion.

for $t := 1$ to T **do**

  for $n := 1$ to N **do**

    if SU n is allowed to transmit at time slot t **then**

      SU n makes a decision based on the current information and then updates its values of z^n, Θ^n, and $\widetilde{\psi}^n$ as follows:

$$z_{k+1}^{n} = \begin{cases} \frac{\nabla \mu^{n}_{\Theta_k^{n}}(s_k^{n},a_k^{n})}{\mu_{\Theta_k^{n}}(s_k^{n},a_k^{n})}, & \text{if } s_k^{n}=s_n^*, \\ z_k^{n} + \frac{\nabla \mu^{n}_{\Theta_k^{n}}(s_k^{n},a_k^{n})}{\mu^{n}_{\Theta_k^{n}}(s_k^{n},a_k^{n})}, & \text{otherwise,} \end{cases} \tag{44}$$

$$\Theta_{k+1}^{n} = \Theta_k^{n} + \rho_k(\mathscr{T}^{n}(s_k^{n},a_k^{n}) - \widetilde{\psi}_k^{n})z_{k+1}^{n}, \tag{45}$$

$$\widetilde{\psi}_{k+1}^{n} = \widetilde{\psi}_k^{n} + \kappa\rho_k(\mathscr{T}^{n}(s_k^{n},a_k^{n}) - \widetilde{\psi}_k^{n}). \tag{46}$$

    end if

    SU n updates information of the data queue and energy level based on environment parameters, e.g., arriving packets and energy harvested.

  end for

end for

VI. Performance Evaluation

A. Parameter Setting

We perform simulations using MATLAB to evaluate the performance of the CRN under different parameter settings and scenarios. First, we consider the scenario with one SU under the attacks from jammers. The capacity of the data queue of the SU is 1 packet, and the capacity of the battery of the SU is 10 units of energy. The packet arrival probability is 0.1. If the data queue is full, an arriving packet will be discarded. If a packet in the data queue is not transmitted successfully, it will be retransmitted until it is successfully received. We assume that the SU uses three units of energy to transmit a packet, while it spends one unit of energy to perform deception, i.e., transmit a fake packet. The jammers need 5 units of energy to perform the attack. The probability that the SU and jammers can harvest one unit of energy is 0.5 in each time slot. There are two strategies for jammers

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

as discussed in Section III, i.e., independent and coordinated attacks. The idle channel probability is 0.85. Furthermore, the successful packet transmission probability by the SU (without jamming) is 0.95 and the sensing error probabilities of the SU, i.e., false alarm and miss detection, are 0.01.

In the case with multiple SUs, we vary the number of SUs and the number of jammers. We compare the performance of the proposed TDMA-learning algorithm with other algorithms, i.e., conventional TDMA, random policy, and a backoff algorithm as presented in [35].

- *Conventional TDMA*: In this algorithm, SUs are allowed to access the channel in a round-robin fashion. In each time slot, if an SU is allowed to use the channel and the channel is idle, it will transmit data if it has a packet in the data queue and enough energy to transmit.
- *Random policy*: When an SU has a packet and enough energy, and the channel is idle, it will select one of two actions, i.e., to transmit data or do nothing, with the same probability, i.e., 0.5.
- *Static backoff algorithm*: If the channel is idle, SUs will transmit data if they have packets and enough energy. If a collision happens, the SUs will select a backoff counter randomly between 1 and 3 and then after each time slot, the backoff counter will be decreased by one. When the backoff counter reaches to zero, the SUs will retransmit data.

For the learning algorithm, we set the parameter vector $\Theta = \mathbf{0}$, i.e., the SU will select an action with uniform probability at the beginning of the algorithm.

B. Simulation Results

1) Single Secondary User - Convergence: We first show the convergence and the optimal policy obtained by the proposed learning algorithm, i.e., Algorithm 2, for the SU with 3 jammers. In Fig. 2, we show the convergence through the average throughput of the SU under two different attack strategies of jammers, i.e., independent and coordinated attacks. As shown in Fig. 2, the convergence of the proposed learning algorithm is relatively fast within around 10^5 iterations. In particular, the average throughput of the learning algorithm converges to 0.04 when the jammers attack independently and the average throughput is approximately 0.0122 when the jammers cooperate to attack. Compared with the greedy strategies of the SU, i.e., the SU will transmit data once it has a packet and enough energy, the average throughput of the proposed algorithm is nearly twice higher.

2) Single Secondary User - Optimal Policy: We then consider the optimal policy obtained from Algorithm 2 under independent and coordinated attacks by jammers, i.e., Fig. 3 and Fig. 4, respectively. In Fig. 3, when the data queue is empty, the SU will do nothing when the energy level is low, and it tends to perform deception as the energy level increases. Additionally, when the data queue is not empty, the probability of doing nothing decreases. In particular, when the energy level is lower than or equal to 3 units, the SU mainly does nothing. By contrast, when the energy level is between 3 and 5 units, the probability of doing nothing is around 0.5 and it reduces

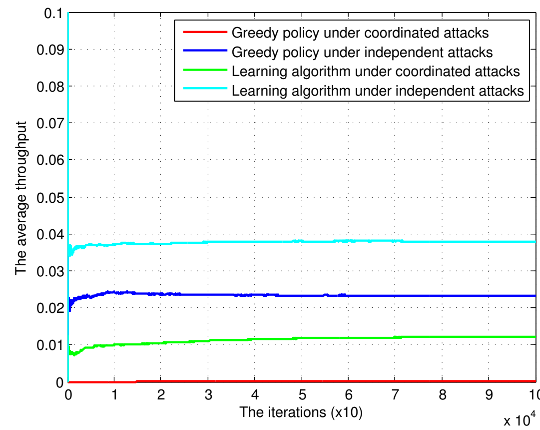


Fig. 2. The convergence of the learning algorithm (3 jammers) under different attack strategies of jammers.

to around 0.25 when the energy level is higher than 6 units. Furthermore, the probabilities to perform deception and data transmission increase as the energy level increases. They are around 0.25 when the energy level is between 3 and 5 units and over 0.30 when the energy level is higher than 5 units.

Similar result is observed when the jammers cooperate to attack as shown in Fig. 4. However, there is a difference at the deception strategy. When the jammers change their strategies from independent to coordinated attack, the efficiency of the attacks will be improved and thus the number of attacks increases. Therefore, in the case of coordinated attacks, the SU has to perform more deception to avoid the attacks from jammers thereby resulting in more opportunities to transmit data successfully. This can be seen clearly in Fig. 5. In Fig. 5, we show the performance of the SU in terms of the ratio of selecting actions and we can observe the probability of performing deception increases from 0.16 to 0.25 when the SU deals with independent and coordinated attacks from jammers, i.e., Fig. 5 (a) and (b), respectively. In addition, the probability of successful deception increases from around 0.75 to more than 0.80 when the jammers's strategies change from independent to coordinated attacks as shown in Fig. 5 (c). Here, we note that the ratio shown in Fig. 5 (a) and (b) are calculated as the number of certain actions chosen by the SU divided by the total number of total actions taken. The successful deception probability in Fig. 5 (c) is defined by the number of times that the SU performs deception and the jammers attack divided by the total number of times that the SU chooses the deception strategy.

3) Single Secondary User - Impact of Energy Harvesting: Next, we vary the energy harvesting probability of the SU and jammers, and evaluate the performance of the system in terms of average throughput of the SU and the average delay of data transmission. To have fair evaluations, we assume that the energy harvesting probabilities of the SU and jammers are the same. We vary the probability as shown in Fig. 6. As the energy harvesting probability of the SU and jammers increases, the average throughput of the SU increases, while the average delay will be reduced for both the learning algorithm and the greedy policy. This is due to the

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

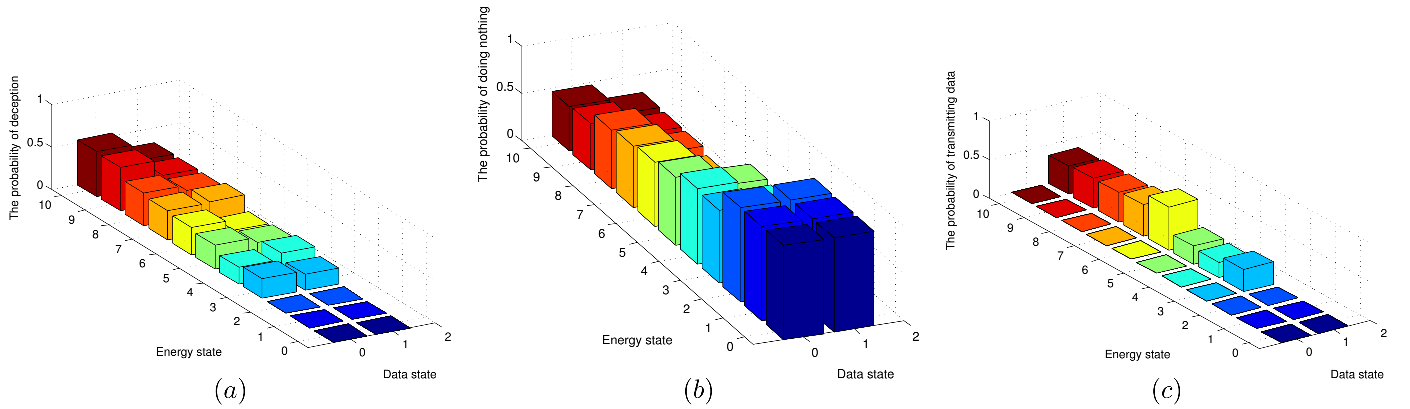


Fig. 3. The optimal policy of the SU under independent attacks. The probability of choosing (a) performing deception, (b) doing nothing, and (c) transmitting data.

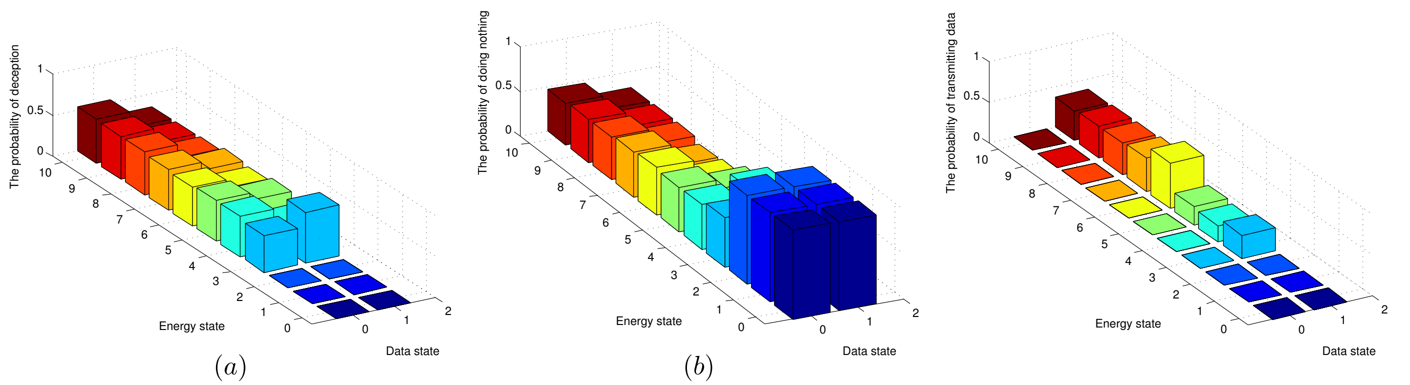


Fig. 4. The optimal policy of the SU under coordinated attacks. The probability of choosing (a) performing deception, (b) doing nothing, and (c) transmitting data.

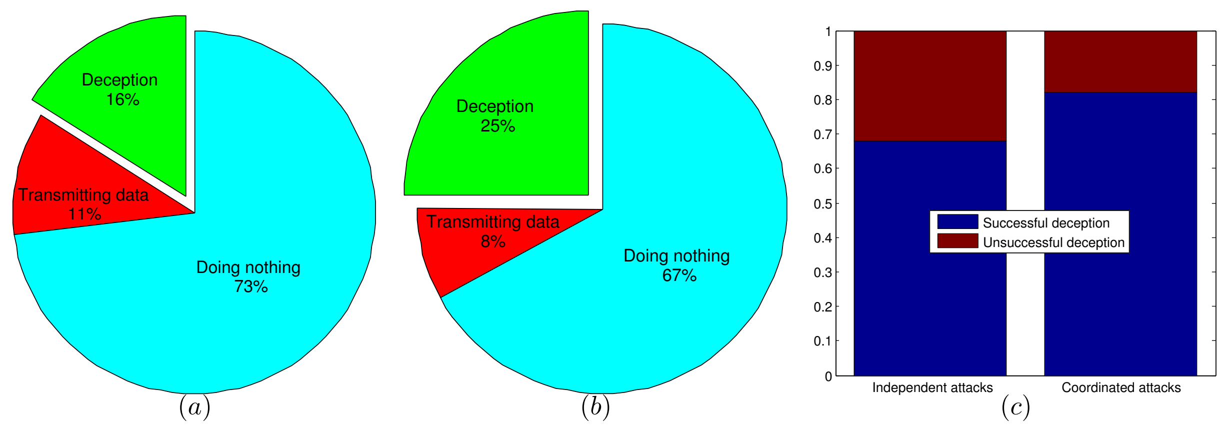


Fig. 5. The SU's performance under (a) independent attacks and (b) coordinated attacks, and (c) the ratio of successful deception.

effective deception mechanism. Additionally, the performance of the SU obtained by the learning algorithm is much better than that of the greedy policy especially when the jammers employ the coordinated attacks. In particular, in Fig. 6 (a), when the attacks are independent, as the energy harvesting probability increases the gap between the average throughput obtained by the learning algorithm and the greedy policy also increases. When the energy harvesting probability is 0.8, the average throughput obtained by the learning algorithm is approximately 1.3 times higher than that of the greedy policy. However, when the energy harvesting probability is 1, the average throughput of both algorithms increases significantly. The reason can be found in Fig. 6. When the jammers are able to harvest energy in every time slot, they will have the same energy levels and the same decisions when they attack the SU. Consequently, the attacks from the jammers are redundant if they apply the independent attack strategy. Thus, the effectiveness of the attack is reduced significantly which makes favorable conditions for the SU to improve its throughput. However, with the coordinated attack strategy, the jammers will not perform unnecessary attacks. Thus, the average throughput of the SU reduces remarkably when the

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

energy harvesting probability is high. With the greedy policy, under the coordinated attacks, its average throughput is nearly zero. In this case, every time when the SU transmits data, there is at least one jammer that is able to attack the SU. However, with the proposed learning algorithm, even with the coordinated attacks, the SU still has an opportunity to transmit packets successfully through using the deception strategy.

4) Multiple Secondary Users - Impact of Number of Secondary Users and Jammers: We now consider the case with multiple SUs coexisting in the same cognitive radio network with jammers. We evaluate the performance in terms of the average throughput of the system when the number of jammers is varied, and the number of SUs is fixed at 5 as shown in Fig. 7. From Fig. 7 (a), as the number of jammers increases, the number of attacks will increase, thus the average throughput from the TDMA-learning algorithm decreases gradually. However, the TDMA-learning algorithm still yields approximately 1.5 times higher throughput than that of the random policy and static backoff algorithm. Here, we observe that the conventional TDMA achieves similar throughput to that of the TDMA-learning. This is due to the fact that when the number of SUs is large, i.e., 5 SUs, using TDMA can avoid collision and the SUs have more time to harvest energy. Consequently, when the SU is scheduled for transmission, mostly it has enough energy to transmit data. Similar results are also observed in Fig. 7 (b) when the jammers perform coordinated attacks. However, in this case, when the number of jammers is large, i.e., double of that of SUs, the average throughput of all algorithms is nearly zero since the jammers now can effectively perform the attacks in every time slot.

Next, we increase the number of SUs while fixing the number of jammers at 5 (as shown in Fig. 8). The average throughput obtained by the TDMA-learning algorithm is still the highest. The proposed TDMA-learning algorithm is patently superior to other algorithms when the number of attacks is many as shown in Fig. 8 (b), i.e., for coordinated attacks. Here, unlike the conventional TDMA and TDMA-learning algorithm, as the number of SUs increases, the average throughput achieved by the random policy and the static backoff algorithm decreases when the number of SUs is greater than 5. The reason is because by using the static backoff algorithm and the random policy, when the number of SUs is large, they will cause more collisions among SUs, thereby reducing the network throughput of SUs. Note that in Fig. 8 we also show the performance of the centralized learning algorithm, i.e., Algorithm 3, when the number of SUs is 2. Clearly, the performance is close to or slightly higher than that of the TDMA-learning algorithm. However, when the number of SUs is greater than 2, we are unable to use centralized learning algorithm because the state space is too large.

Finally, we consider the impact of sensing errors of jammers to the performance of the system. The wrong sensing results of jammers appear in four cases. First, when the channel is idle and there is at least one SU transmitting data to the channel, but the jammer does not attack because it cannot detect signals of SUs. Second, when the channel is idle and there is no SU transmitting data, but the jammer still attacks the channel because there are some noise signals which are similar to secondary users' signals. Third, the channel is busy and there is at least one SU transmitting data (due to the wrong sensing result of SUs), but the jammer does not attack because it cannot detect SU's signals. Fourth, the channel is busy and there is no SU transmitting data, but the jammer still attacks the channel (e.g., jammers misinterpret that PUs' signals are SU's signals because of wrong sensing result). In such four cases, the third case will bring the benefit for jammer, while the other cases will have negative impacts to the jammers. In particular, in the first case, the jammer will not attack when SUs transmit data, while in the second and fourth cases, the jammer will attack the channel when there is no SUs' transmission. This will undermine the efficiency of attacks by the jammers, while enhancing the network performance for SUs. This is verified through simulation results obtained in Fig. 9, i.e., as the sensing error probability of jammers increases, the throughput of the system will also increase.

In summary, we make two important observations as follows:

- In the case with one secondary user, the learning algorithm (i.e., Algorithm 2) yields much better performance than that of the greedy policy. Additionally, with the learning algorithm, the secondary user can automatically and adaptively adjust its strategy according to different strategies of jammers and the dynamics of the cognitive radio environment.
- In the case with more than one secondary user, the TDMA-learning algorithm yields the results close to that obtained by the centralized learning algorithm. Moreover, the results obtained by the TDMA-learning algorithm are much better than that of other algorithms, i.e., conventional TDMA, random policy and backoff algorithm under different numbers of jammers and their strategies.

VII. Conclusion

In this paper, we have presented the cognitive radio network that allows secondary users to harvest energy. The secondary users can receive energy from wireless chargers and then use this energy to transmit data on a channel. To deal with jamming attacks as well as unknown information of the environment, we have proposed a learning algorithm developed from the simulation-based and policy gradient methods. The algorithm helps the secondary users not only to deal with different jammers' strategies, but also to adjust their data transmission or deception in an online fashion to achieve the maximum throughput. We have also extended the optimization problem with multiple secondary users and proposed the learning algorithms to help secondary users maximize the overall throughput. Through simulations, we have shown the convergence as well as the efficiency of the proposed learning algorithms compared with other conventional and static algorithms. In the future work, we will consider applying the deception strategy along with the online learning algorithm for cognitive radio networks with multiple channels and

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

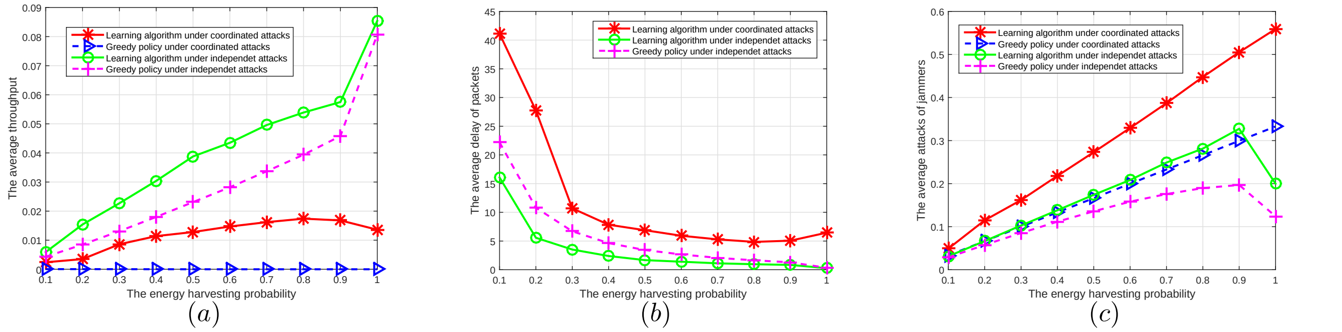


Fig. 6. (a) The average throughput and (b) the average delay of the SU, and (c) the average attacks of jammers, when the energy harvesting probability of the SU and jammers is varied.

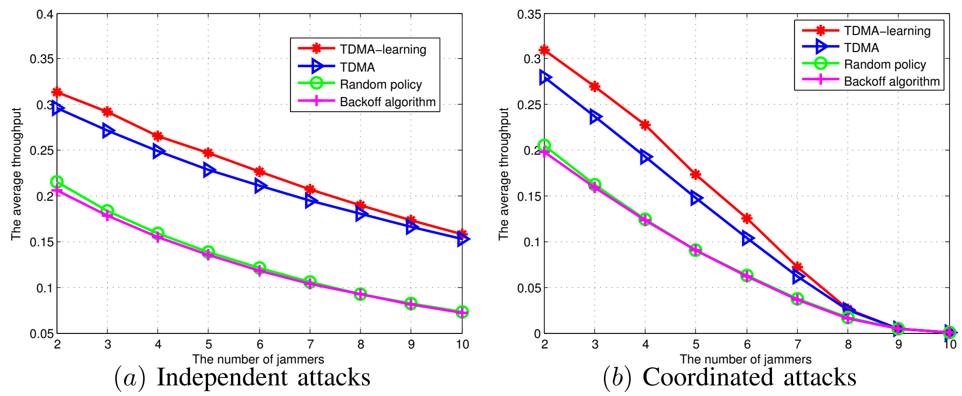


Fig. 7. Average throughput of the system under (a) independent and (b) coordinated attacks by the jammers when the number of jammers is varied.

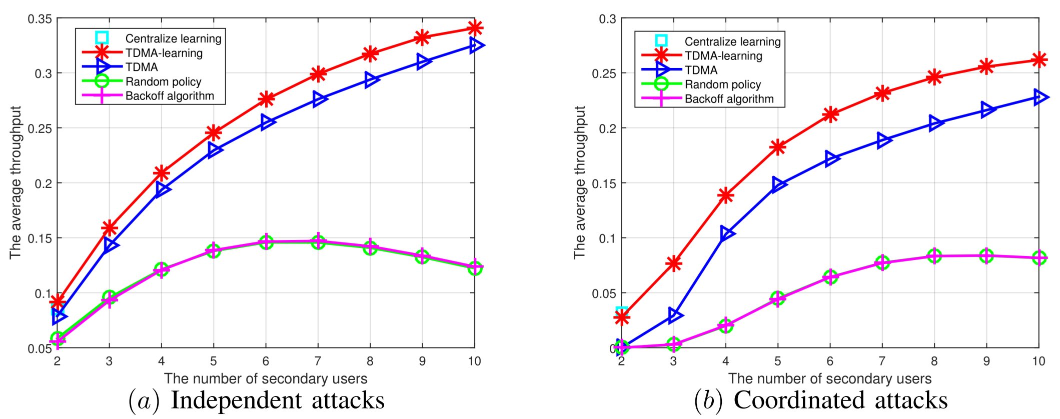


Fig. 8. Average throughput of the system under (a) independent and (b) coordinated attacks from jammers when the number of secondary users is varied

compare the network performance with other channel hopping strategies.

Appendix A
The proof of Proposition 1

This is to show the gradient of the average throughput. In (12), we have $\sum_{s\in\mathcal{S}}\pi_\Theta(s)=1$, so $\sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)=0$.

Recall that

$$d(s,\Theta)=\mathscr{T}(s,\Theta)-\psi(\Theta)+\sum_{s'\in\mathcal{S}}\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta)$$

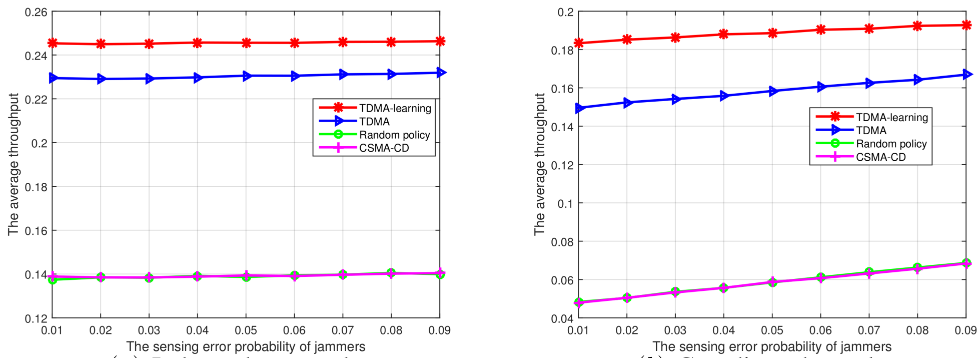
$$\text{and}\quad \psi(\Theta)=\sum_{s\in\mathcal{S}}\pi_\Theta(s)\mathscr{T}(s,\Theta).$$

Then, we derive the following results:
We define

$$\begin{aligned}&\nabla\Big(\pi_\Theta(s)\mathbf{p}(s'|s,\Psi(\Theta))\Big)=\\&\nabla\pi_\Theta(s)\mathbf{p}(s'|s,\Psi(\Theta))+\pi_\Theta(s)\nabla\mathbf{p}(s'|s,\Psi(\Theta)),\end{aligned}\tag{50}$$

and from (12), $\sum_{s\in\mathcal{S}}\pi_\Theta(s)\mathbf{p}(s'|s,\Psi(\Theta))=\pi_\Theta(s'),$ for $s'\in$

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking



(a) Independent attacks (b) Coordinated attacks

Fig. 9. Average throughput of the system under (a) independent and (b) coordinated attacks by the jammers when the sensing error probability of jammers is varied.

$$\nabla\psi(\Theta) = \sum_{s\in\mathcal{S}}\pi_\Theta(s)\nabla\mathscr{T}(s,\Theta) + \sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)\mathscr{T}(s,\Theta), \tag{47}$$

$$= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\nabla\mathscr{T}(s,\Theta) + \sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)\mathscr{T}(s,\Theta) - \psi(\Theta)\sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s), \tag{48}$$

$$= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\nabla\mathscr{T}(s,\Theta) + \sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)\big(\mathscr{T}(s,\Theta) - \psi(\Theta)\big), \tag{49}$$

$$= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\nabla\mathscr{T}(s,\Theta) + \sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)\Big(d(s,\Theta) - \sum_{s'\in\mathcal{S}}\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta)\Big).$$

$$\begin{aligned}\nabla\psi(\Theta) &= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\nabla\mathscr{T}(s,\Theta) + \sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)\Big(d(s,\Theta) - \sum_{s'\in\mathcal{S}}\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta)\Big),\\ &= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\nabla\mathscr{T}(s,\Theta) + \sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)d(s,\Theta)\\ &\quad + \sum_{s,s'\in\mathcal{S}}\Big(\pi_\Theta(s)\nabla\mathbf{p}(s'|s,\Psi(\Theta)) - \nabla\Big(\pi_\Theta(s)\mathbf{p}(s'|s,\Psi(\Theta))\Big)\Big)d(s',\Theta), \qquad (51)\\ &= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\nabla\mathscr{T}(s,\Theta) + \sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)d(s,\Theta)\\ &\quad + \sum_{s,s'\in\mathcal{S}}\pi_\Theta(s)\nabla\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta) - \sum_{s'\in\mathcal{S}}\nabla\Big(\sum_{s\in\mathcal{S}}\pi_\Theta(s)\mathbf{p}(s'|s,\Psi(\Theta))\Big)d(s',\Theta), \qquad (52)\\ &= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\nabla\mathscr{T}(s,\Theta) + \sum_{s\in\mathcal{S}}\nabla\pi_\Theta(s)d(s,\Theta)\\ &\quad + \sum_{s,s'\in\mathcal{S}}\pi_\Theta(s)\nabla\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta) - \sum_{s'\in\mathcal{S}}\nabla\pi_\Theta(s')d(s',\Theta), \qquad (53)\\ &= \sum_{s\in\mathcal{S}}\pi_\Theta(s)\Big(\nabla\mathscr{T}(s,\Theta) + \sum_{s'\in\mathcal{S}}\nabla\mathbf{p}(s'|s,\Psi(\Theta))d(s',\Theta)\Big). \qquad (54)\end{aligned}$$

$\mathcal{S}$. Then, we have the derivations as given in (51)-(54). The proof is completed.

Appendix B
The proof of Proposition 2

We prove the convergence of the Algorithm 1. The update equations of Algorithm 1 can be rewritten in the specific form as in (55).

We define the vector $\mathbf{r}^{k_m} = \begin{bmatrix} \Theta_m & \widetilde{\psi}_m \end{bmatrix}^\top$, then (55) becomes

$$\mathbf{r}^{k_{m+1}} = \mathbf{r}^{k_m} + \rho_m\mathbf{H}_m, \tag{56}$$

where

$$\mathbf{H}_m = \begin{bmatrix} \sum_{k'=k_m}^{k_{m+1}-1}\Big(\sum_{k=k'}^{k_{m+1}-1}(\mathscr{T}(s_k,a_k) - \widetilde{\psi}_m)\Big)\frac{\nabla\mu_{\Theta_m}(s_{k'},a_{k'})}{\mu_{\Theta_m}(s_{k'},a_{k'})} \\ \kappa\sum_{k'=k_m}^{k_{m+1}-1}(\mathscr{T}(s_k,a_k) - \widetilde{\psi}_m) \end{bmatrix}.$$

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2015.2488620, IEEE Transactions on Cognitive Communications and Networking

$$\begin{aligned}
\Theta_{m+1} &= \Theta_m + \rho_m \left(\sum_{k'=k_m}^{k_{m+1}-1} \Big(\sum_{k=k'}^{k_{m+1}-1} (\mathscr{T}(s_k, a_k) - \widetilde{\psi}_m) \Big) \frac{\nabla \mu_{\Theta_m}(s_{k'}, a_{k'})}{\mu_{\Theta_m}(s_{k'}, a_{k'})} \right), \\
\widetilde{\psi}_{m+1} &= \widetilde{\psi}_m + \kappa \rho_m \sum_{k'=k_m}^{k_{m+1}-1} (\mathscr{T}(s_k, a_k) - \widetilde{\psi}_m)
\end{aligned} \tag{55}$$

Let $\mathscr{F} = \{\Theta_0, \widetilde{\psi}_0, s_0, s_1, \dots, s_m\}$ be the history of the Algorithm 1. Then from Proposition 2 in [30], we have

$$\mathbb{E}[\mathbf{H}_m|\mathscr{F}_m] = \mathbf{h}_m = \begin{bmatrix} \mathbb{E}_\Theta[T]\nabla\psi(\Theta) + \mathscr{V}(\Theta)\big(\psi(\Theta) - \widetilde{\psi}(\Theta)\big) \\ \kappa \mathbb{E}_\Theta[T]\big(\psi(\Theta) - \widetilde{\psi}(\Theta)\big) \end{bmatrix},$$

where

$$\mathscr{V}(\Theta) = \mathbb{E}_\Theta \left[\sum_{k'=k_{m+1}}^{k_{m+1}-1} \big(k_{m+1} - k'\big) \frac{\nabla \mu_{\Theta_m}(s_{k'}, a_{k'})}{\mu_{\Theta_m}(s_{k'}, a_{k'})} \right].$$

Recall that $T = \min\{k > 0 | s_k = s^*\}$ is the first future time that the recurrent state s^* is visited.

Consequently, (55) has the following form

$$\mathbf{r}^{k_{m+1}} = \mathbf{r}^{k_m} + \rho_m \mathbf{h}_m + \varepsilon_m, \tag{57}$$

where $\varepsilon_m = \rho(\mathbf{H}_m - \mathbf{h}_m)$ and note that $\mathbb{E}[\varepsilon_m|\mathscr{F}_m] = 0$. Since ε_m and ρ_m converge to zero almost surely, along with $\mathbf{h}_m$ is bounded, we have

$$\lim_{m\to\infty} (\mathbf{r}^{k_{m+1}} - \mathbf{r}^{k_m}) = 0. \tag{58}$$

After that, based on Lemma 11 in [30], it is proved that $\psi(\Theta)$ and $\widetilde{\psi}(\Theta)$ converge to a common limit. This means that the parameter vector Θ can be represented as follows:

$$\Theta_{m+1} = \Theta_m + \rho_m \mathbb{E}_{\Theta_m}[T](\nabla\psi(\Theta_m) + e_m) + \epsilon_m, \tag{59}$$

where e_m is an error term that converges to zero and ϵ_m is a summable sequence. (59) is known as the gradient method with diminishing errors [36], [37]. Therefore, following the same way in [36], [37], we can prove that $\nabla\psi(\Theta_m)$ converges to 0, i.e., $\nabla_\Theta \psi(\Theta_\infty) = 0$.

Appendix C
The proof of Proposition 3

Assume that at time slot t, the system is at state $s^{\text{cen}} = (s^1, \dots, s^N)$ and the action space corresponding to the current state is $\mathcal{A}^{\text{cen}}_{s^{\text{cen}}} = (\mathcal{A}^1_{s^1}, \dots, \mathcal{A}^N_{s^N})$. We denote $|\mathcal{A}^n_{s^n}|$ as the cardinality of the action space $\mathcal{A}^n$ of secondary user n given the current state s_n. Then, we note an important feature in the system that there is only one secondary user using the channel at a time. Thus, if any secondary user selects action 2 or 3 (i.e., deception or transmit data), all the other secondary users must select action 1, i.e., to do nothing. Additionally, we note that when $\mathcal{A}^n = 1$, the secondary user n has only one action, i.e., 1. When $\mathcal{A}^n = 2$, the secondary user n has two actions to choose, i.e., 1 and 2. When $\mathcal{A}^n = 3$, the secondary user n has three actions, i.e., 1, 2 and 3. Hence, from aforementioned property, we derive the following results. In (60), when $\mathcal{A}^n_{s^n} \geq 2$, we will count the number of actions that is larger than or equal to 2. This is because $\mathcal{A}^n_{s^n} \geq 1, \forall n$ and if secondary user n selects action 2 or 3, then all other secondary users will select action 1. If secondary user n selects action 1, then we perform the counting process for the rest of users. The proof is completed.

References

[1] E. Hossain, D. Niyato, and Z. Han, *Dynamic Spectrum Access and Management in Cognitive Radio Networks*, Cambridge University Press, Cambridge, UK. 2009.

[2] Y. Liu, P. Ning, and H. Dai, "Authenticating primary users' signals in cognitive radio networks via integrated cryptographic and wireless link signatures," in *IEEE Symposium on Security and Privacy*, pp. 286-301, Oakland, USA, May. 2010.

[3] D. Pu, "Primary user emulation detection in cognitive radio networks," Ph.D. dissertation, Dept. of Electrical & Computer Engineering, Worcester Polytechnic Institute, Worcester Massachusetts, USA. 2013.

[4] R. Chen, J.-M. Park, and J. H. Reed, "Defense against primary users emulation attacks in cognitive radio networks," *IEEE Journal on Selected Areas in Communications*, vol. 26, no. 1, pp. 25-37, Jan. 2008.

[5] A. Sultan, "Sensing and transmit energy optimization for an energy harvesting cognitive radio," *IEEE Wireless Communications Letters*, vol. 1, no. 5, pp. 500-503, Oct. 2012.

[6] X. Gao, W. Xu, S. Li, and J. Lin, "An online energy allocation strategy for energy harvesting cognitive radio systems," in *International Conference on Wireless Communications & Signal Processing*, Hangzhou, China, pp. 1-5, Oct. 2013.

[7] S. Park, H. Kim, and D. Hong, "Cognitive radio networks with energy harvesting," *IEEE Transactions on wireless communications*, vol. 12, no. 3, pp. 1386-1397, Mar. 2013.

[8] Q. Zhang, B. Cao, Y. Wang, N. Zhang, X. Lin, and L. Sun, "On exploiting polarization for energy-harvesting enabled cooperative cognitive radio networking," *IEEE Wireless Communications*, vol. 20, no. 4, pp. 116-124, Aug. 2013.

[9] S. Park, J. Heo, B. Kim, and W. Chung, "Optimal mode selection for cognitive radio sensor networks with RF energy harvesting," in *Proceedings of IEEE International Symposium on Personal Indoor and Mobile Radio Communications*, pp. 2155-2159, Sydney, Australia, Sept. 2012.

[10] S. Park and D. Hong, "Optimal spectrum access for energy harvesting cognitive radio networks," *IEEE Transactions on Wireless Communications*, vol. 12, no. 12, pp. 6166-6179, Dec. 2013.

[11] D. T. Hoang, D. Niyato, P. Wang, and D. I. Kim, "Opportunistic channel access and RF energy harvesting in cognitive radio networks," *IEEE Journal of Selected Areas in Communications*, vol. 32, no. 11, pp. 2039-2052, Nov. 2014.

[12] D. T. Hoang, D. Niyato, P. Wang, and D. I. Kim, "Performance optimization for cooperative multiuser cognitive radio networks with RF energy harvesting capability ," *IEEE Transactions on Wireless Communications*, accepted for final publication, Feb. 2015.

[13] A. Kurs, A. Karalis, R. Moffatt, J. D. Joannopoulos, P. Fisher, and M. Soljacic, "Wireless power transfer via strongly coupled magnetic resonances," *Science*, vol. 317, no. 5834, pp. 83-86, Jul. 2007.

[14] P. Nintanavongsa, U. Muncuk, D. Richard Lewis, and K. R. Chowdhury, "Design optimization and implementation for RF energy harvesting circuits," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 2, no. 1, pp. 24-33, Mar. 2012.

[15] H. Ostaffe, "Power out of thin air: Ambient RF energy harvesting for wireless sensors," 2010 [Online] powercastco.com/PDF/Power-Out-of-Thin-Air.pdf.

[16] H. Li and Z. Han, "Dogfight in spectrum: Combating primary user emulation attacks in cognitive radio systems Part I: Known channel statistics," *IEEE Transactions on Wireless Communications*, vol.9, no.11, pp. 3566-3577, Nov. 2010.

$$|\mathcal{A}^{\mathrm{cen}}_{s^{\mathrm{cen}}}| = \underbrace{(|\mathcal{A}^1_{s^1}|-1)}_{|\mathcal{A}^1_{s^1}|\geq 2} + \underbrace{\Bigg[\underbrace{(|\mathcal{A}^2_{s^2}|-1)}_{|\mathcal{A}^2_{s^2}|\geq 2} + \underbrace{\Bigg[\underbrace{(|\mathcal{A}^3_{s^3}|-1)}_{|\mathcal{A}^3_{s^3}|\geq 2} + \underbrace{\cdots + \Big[\underbrace{(|\mathcal{A}^n_{s^n}|-1)}_{|\mathcal{A}^n_{s^n}|\geq 2} + \underbrace{1}_{|\mathcal{A}^n_{s^n}|=1}\Big]}_{|\mathcal{A}^3_{s^3}|=1}\Bigg]}_{|\mathcal{A}^2_{s^2}|=0}\Bigg]}_{|\mathcal{A}^1_{s^1}|=1}, \quad (60)$$

$$= \sum_{n=1}^{N} |\mathcal{A}^n_{s^n}| - N + 1. \quad (61)$$

[17] H. Li and Z. Han, "Dogfight in spectrum: Combating primary user emulation attacks in cognitive radio systems Part II: Unknown channel statistics," *IEEE Transactions on Wireless Communications*, vol.10, no.1, pp. 274-283, Jan. 2011.

[18] P. Auer, N. Cesa-Bianchi, Y. Freund, and R. E. Schapire, "The nonstochastic multi-armed bandit problem," *SIAM Journal on Computing*, vol. 32, issue 1, pp. 48-77, Jan. 2003.

[19] S. Sodagari and T. C. Clancy, "An anti-jamming strategy for channel access in cognitive radio networks," in *Proceedings of the Second international conference on Decision and Game Theory for Security*, pp. 34-43, Berlin, Germany, Nov. 2011.

[20] J. Vermorel and M. Mohri, "Multi-armed bandit algorithms and empirical evaluation," in *Proceedings of the 16th European conference on Machine Learning*, pp. 437-448, Berlin, Germany, Oct. 2005.

[21] Y. Wu, B. Wang, K. J. R. Liu, and T. C. Clancy, "Anti-jamming games in multi-channel cognitive radio networks," *IEEE Journal on Selected Areas in Communications*, vol. 30, no.1, pp. 4-15, Jan. 2012.

[22] B. Wang, Y. Wu, K. J. R. Liu, and T. C. Clancy, "An anti-jamming stochastic game for cognitive radio networks," *IEEE Journal on Selected Areas in Communications*, vol. 29, no.4, pp. 877-889, Apr. 2011.

[23] W. Wang, S. Bhattacharjee, M. Chatterjee, and K. Kwiat, "Collaborative jamming and collaborative defense in cognitive radio networks," *Pervasive and Mobile Computing*, vol. 9, issue 4, pp. 572-587, Aug. 2013.

[24] S. Bhunia, X. Su, S. Sengupta, and F. Vazquez-Abad, "Stochastic model for cognitive radio networks under jamming attacks and honeypot-based prevention," in *International Conference on Distributed Computing and Networks*, pp. 438-452, Berlin, Germany, Jan. 2014.

[25] Y. Wu, B. Wang, and K. J. R. Liu, "Optimal power allocation strategy against jamming attacks using the Colonel Blotto game," in *IEEE Global Telecommunications Conference*, pp. 1-5, Hawaii, USA, Dec. 2009.

[26] B. Roberson, "The Colonel Blotto game," *Economic Theory*, vol. 29, no. 1, pp. 1-24, Sept. 2006.

[27] D. Fudenburg and J. Tirole, *Game Theory*, MIT Press, 1993.

[28] C. D. Schleher, "Electronic Warfare in the Information Age," Norwood, Artech House. 1999.

[29] K. Pelechrinis, M. Iliofotou, and S. V. Krishnamurthy, "Denial of service attacks in wireless networks: The case of jammers," *IEEE Communications Surveys and Tutorials*, vol. 13, no. 2, pp. 245-257, May. 2011.

[30] P. Marbach and J. N. Tsitsiklis, "Simulation-based optimization of Markov reward processes," *IEEE Transactions on Automatic Control*, vol. 46, issue 2, pp. 191-209, Feb. 2001.

[31] O. Buffet, A. Dutech, and F. Charpillet, "Shaping multi-agent systems with gradient reinforcement learning," *Journal of Autonomous Agents and Multi-Agent System*, vol. 15, issue 2, pp. 197-220, Jan. 2007.

[32] J. Baxter, P. L. Barlett, and L. Weaver, "Experiments with infinite-horizon, policy-gradient estimation," *Journal of Artificial Intelligence Research*, vol. 15, issue 1, pp. 351-381, Nov. 2001.

[33] Abhijit Gosavi, *Simulation-Based Optimization: Parametric Optimization Techniques and Reinforcement Learning*, Kluwer Academic Publishers Norwell, MA, USA. 2003.

[34] Dimitri P. Bertsekas, "Nonlinear Programming," Athena Scientific, Belmont, MA, 1995.

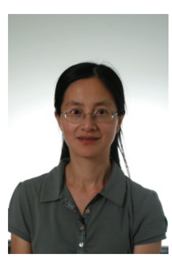
[35] D. T. Hoang, and D. Niyato, "Performance analysis for cognitive machine-to-machine communications," in *IEEE International Conference on Communication Systems*, pp. 245-249, Singapore, Nov. 2012.

[36] D. P. Bertsekas, and J. N. Tsitsiklis, "Gradient convergence in gradient methods with errors," in *SIAM Journal on Optimization*, vol. 10, issue 3, pp. 627-642, 1999.

[37] V. S. Borkar, "Stochastic Approximation: A Dynamic Systems Viewpoint," Cambridge University Press, 2008.

Dinh Thai Hoang (S'15) received his Bachelor degree in Electronics and Telecommunications from Hanoi University of Science and Technology (HUST), Vietnam in 2009. From 2010 to 2012 he worked as a research assistant at Nanyang Technological University (NTU), Singapore. He is currently working toward his Ph.D. degree in School of Computer Engineering, NTU under Professor Dusit Niyato's supervision. His research interests include optimization problems for wireless communication networks.

Dusit Niyato (M'09-SM'15) is currently an Associate Professor in the School of Computer Engineering, at Nanyang Technological University, Singapore. He received B.E. from King Mongkuts Institute of Technology Ladkrabang (KMITL) in 1999. He obtained his Ph.D. in Electrical and Computer Engineering from the University of Manitoba, Canada in 2008. His research interests are in the area of radio resource management in cognitive radio networks and energy harvesting for wireless communication.



Ping Wang (M'08-SM'15) received the PhD degree in electrical engineering from University of Waterloo, Canada, in 2008. She is currently an Associate Professor in the School of Computer Engineering, Nanyang Technological University, Singapore. Her current research interests include resource allocation in multimedia wireless networks, cloud computing, and smart grid. She was a corecipient of the Best Paper Award from IEEE Wireless Communications and Networking Conference (WCNC) 2012 and IEEE International Conference on Communications (ICC) 2007. She is an Associate Editor of IEEE Transactions on Wireless Communications, EURASIP Journal on Wireless Communications and Networking, and International Journal of Ultra Wideband Communications and Systems.

Dong In Kim (S'89-M'91-SM'02) received the Ph.D. degree in electrical engineering from the University of Southern California, Los Angeles, CA, USA, in 1990. He was a tenured Professor with the School of Engineering Science, Simon Fraser University, Burnaby, BC, Canada. Since 2007, he has been with Sungkyunkwan University (SKKU), Suwon, Korea, where he is currently a Professor with the College of Information and Communication Engineering. Dr. Kim has served as an Editor and a Founding Area Editor of Cross-Layer Design and Optimization for the IEEE Transactions on Wireless Communications from 2002 to 2011. From 2008 to 2011, he served as the Co-Editor-in-Chief for the Journal of Communications and Networks (JCN). He is currently the Founding Editor-in-Chief for the IEEE Wireless Communications Letters and has been serving as an Editor of Spread Spectrum Transmission and Access and an Editor-at-Large in Wireless Communication for the IEEE Transactions on Communications since 2001. He is a first recipient of the NRF of Korea Engineering Research Center (ERC) in Wireless Communications for Energy Harvesting Communications (2014-2021).